04 Limits

1. INTRODUCTION:

The concept of limit of a function is one of the fundamental ideas that distinguishes calculus from algebra and trigonometry. We use limits to describe the way a function f varies. Some functions vary continuously; small changes in x produce only small changes in $f(x)$. Other functions can have values that jump or vary erratically. We also use limits to define tangent lines to graphs of functions. This geometric application leads at once to the important concept of derivative of a function

2. DEFINITION:

Let $f(x)$ be defined on an open interval about 'a' except possibly at 'a' itself. If $f(x)$ gets arbitrarily close to L (a finite number) for all x sufficiently close to 'a' we say that $f(x)$ approaches the limit L as x approaches 'a' and we write $\lim_{x\to a} f(x) = L$ and say "the limit of $f(x)$, as x approaches a, equals L".

This implies if we can make the value of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a.

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3. LEFT HAND LIMIT AND RIGHT HAND LIMIT OF A FUNCTION:

The value to which $f(x)$ approaches, as x tends to 'a' from the left hand side $(x \to a^-)$ is called left hand limit of $f(x)$ at $x = a$. Symbolically, LHL = $\lim_{x\to a^-} f(x) = \lim_{h\to 0} f(a-h)$.

The value to which $f(x)$ approaches, as x tends to 'a' from the right hand side $(x \to a^+)$ is called right hand limit of $f(x)$ at $x = a$. Symbolically, RHL = $\lim_{x\to a^+} f(x) = \lim_{h\to 0} f(a+h)$.

Limit of a function $f(x)$ is said to exist as, $x \to a$ when $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$ = Finite quantity.

Example:

Graph of $y = f(x)$

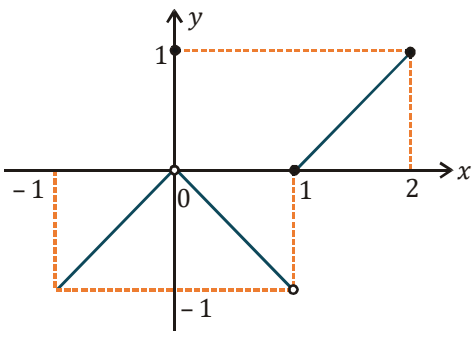


Fig. 1

$\lim_{x\to -1^+} f(x) = \lim_{h\to 0} f(-1+h) = f(-1^+) = -1$

$\lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h) = f(0^-) = 0$

$\lim_{x\to 1^-} f(x) = \lim_{h\to 0} f(1-h) = f(1^-) = -1$

$\lim_{x\to 1^+} f(x) = \lim_{h\to 0} f(1+h) = f(1^+) = 0$

$\lim_{x\to 2^-} f(x) = \lim_{h\to 0} f(2-h) = f(2^-) = 1$

$\lim_{x\to 0} f(x) = 0$ and $\lim_{x\to 1} f(x)$ does not exist.

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Important Note:

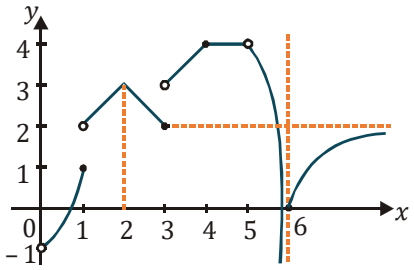
In $\lim_{x\to a} f(x), x \neq a$ $\boldsymbol{x \to a}$ **necessarily implies**. That is while evaluating limit at $x = a$, we are not concerned with the value of the function at $x = a$. In fact, the function may or may not be defined at $x = a$.

Also, it is necessary to note that if $f(x)$ is defined only on one side of '$\boldsymbol{x = a}$', one sided limit are good enough to establish the existence of limits, & if $f(x)$ is defined on either side of '$\boldsymbol{a}$' both sided limits are to be considered.

As in $\lim_{x\to 1} \cos^{-1} x = 0$, though $f(x)$ is not defined for $x > 1$, even in its immediate vicinity.

Illustration 1

Consider the adjacent graph of $y = f(x)$ Find the following:



(a) $\lim_{x\to 0^-} f(x)$ (b) $\lim_{x\to 0^+} f(x)$ (c) $\lim_{x\to 1^-} f(x)$

(d) $\lim_{x\to 1^+} f(x)$ (e) $\lim_{x\to 2^-} f(x)$ (f) $\lim_{x\to 2^+} f(x)$

(g) $\lim_{x\to 3^-} f(x)$ (h) $\lim_{x\to 3^+} f(x)$ (i) $\lim_{x\to 4^-} f(x)$

(j) $\lim_{x\to 4^+} f(x)$ (k) $\lim_{x\to \infty} f(x) = 2$ (l) $\lim_{x\to 6^-} f(x) = -\infty$

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Solution:

(a) As $x \to 0^-$: limit does not exist (the function is not defined to the left of $x = 0$)

(b) As $x \to 0^+ : f(x) \to -1 \Rightarrow \lim_{x\to 0^+} f(x) = -1$.

(c) As $x \to 1^- : f(x) \to 1 \Rightarrow \lim_{x\to 1^-} f(x) = 1$.

(d) As $x \to 1^+ : f(x) \to 2 \Rightarrow \lim_{x\to 1^+} f(x) = 2$.

(e) As $x \to 2^- : f(x) \to 3 \Rightarrow \lim_{x\to 2^-} f(x) = 3$.

(f) As $x \to 2^+ : f(x) \to 3 \Rightarrow \lim_{x\to 2^-} f(x) = 3$.

(g) As $x \to 3^- : f(x) \to 2 \Rightarrow \lim_{x\to 3^-} f(x) = 2$.

(h) As $x \to 3^+ : f(x) \to 3 \Rightarrow \lim_{x\to 3^+} f(x) = 3$.

(i) As $x \to 4^- : f(x) \to 4 \Rightarrow \lim_{x\to 4^-} f(x) = 4$.

(j) As $x \to 4^+ : f(x) \to 4 \Rightarrow \lim_{x\to 4^+} f(x) = 4$.

(k) As $x \to \infty : f(x) \to 2 \Rightarrow \lim_{x\to \infty} f(x) = 2$.

(l) As $x \to 6^-, f(x) \to -\infty \Rightarrow \lim_{x\to 6^-} f(x) = -\infty$ limit does not exist because it is not finite.

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Illustration 2

Let $[t]$ denote the greatest integer $\le t$. If for some $\lambda \in R - \{0, 1\}$, $\lim\limits_{x\to 0}\left|\dfrac{1-x+|x|}{\lambda-x+[x]}\right| = L$, then L is equal to :

(A) 1 (B) 2 (C) $\frac{1}{2}$ (D) 0

Ans. (B)

Solution:

LHL : $\lim\limits_{x\to 0^-}\left|\dfrac{1-x-x}{\lambda-x-1}\right| = \left|\dfrac{1}{\lambda-1}\right|$

RHL : $\lim\limits_{x\to 0^+}\left|\dfrac{1-x+x}{\lambda-x+1}\right| = \left|\dfrac{1}{\lambda}\right|$

For existence of limit

LHL = RHL

$\Rightarrow \dfrac{1}{|\lambda-1|} = \dfrac{1}{|\lambda|} \Rightarrow \lambda = \dfrac{1}{2}$

$\therefore\ L = \dfrac{1}{|\lambda|} = 2$

Illustration 3

Let a be an integer such that $\lim\limits_{x\to 7}\dfrac{18-[1-x]}{[x-3a]}$ exists, where $[t]$ is greatest integer $\le t$. Then a is equal to :

(A) –6 (B) –2 (C) 2 (D) 6

Ans. (A)

Solution:

$\lim\limits_{x\to 7}\dfrac{18-[1-x]}{[x]-3a}$

L.H.L. $\lim\limits_{x\to 7^-}\dfrac{18-[1-x]}{[x]-3a}$

$= \dfrac{18-(-6)}{6-3a}$

$= \dfrac{24}{6-3a}$

R.H.L. $\lim\limits_{x\to 7^+}\dfrac{18-[1-x]}{[x]-3a}$

$= \dfrac{18-(-7)}{7-3a}$

$= \dfrac{25}{7-3a}$

Now L.H.L. = R.H.L.

$\dfrac{24}{6-3a} = \dfrac{25}{7-3a}$

$\Rightarrow 168-72a = 150-75a$

$\Rightarrow 18 = -3a$

$\Rightarrow a = -6$

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Illustration 4

Let $f(x) = \begin{cases} x-1, & x \text{ is even} \\ 2x, & x \text{ is odd,} \end{cases}$ $x \in N$. If for some $a \in N$, $f(f(f(a))) = 21$, then $\lim\limits_{x \to a^-} \left\{ \frac{|x|^3}{a} - \left[\frac{x}{a} \right] \right\}$ where $[t]$ denotes the greatest integer less than or equal to t, is equal to :

(A) 121 (B) 144 (C) 169 (D) 225

Ans. (B)

Solution:

$f(x) = \begin{cases} x-1, & x \text{ is even} \\ 2x, & x \text{ is odd,} \end{cases}$

$f(f(f(a))) = 21$

C–1: If a = even

$f(a) = a - 1$ = odd

$f(f(a)) = 2(a - 1)$ = even

$f(f(f(a))) = 2a - 3 = 21 \Rightarrow a = 12$

C–2: If a = odd

$f(a) = 2a$ = even

$f(f(a)) = 2a - 1$ = odd

$f(f(f(a))) = 4a - 2 = 21$ (Not possible)

Hence $a = 12$

Now

$\lim\limits_{x \to 12^-} \left(\frac{|x|^3}{2} - \left[\frac{x}{12} \right] \right)$

$\lim\limits_{x \to 12^-} \frac{|x|^3}{12} - \lim\limits_{x \to 12^-} \left[\frac{x}{12} \right]$

$= 144 - 0 = 144.$

Illustration 5

Let $\{x\}$ denote the fractional part of x and $f(x) = \frac{\cos^{-1} 1 - \{x\}^2 \sin^{-1}(1 - \{x\})}{\{x\} - \{x\}^3}$, $x \neq 0$. If L and R respectively denotes the left hand limit and the right hand limit of $f(x)$ at x = 0, then $\frac{32}{\pi^2}(L^2 + R^2)$ is equal to _____.

Ans. (18)

Solution:

Finding right hand limit

$\lim\limits_{x \to 0^+} f(x) = \lim\limits_{h \to \infty} f(0 + h)$

$= \lim\limits_{h \to 0} f(h)$

$= \lim\limits_{h \to 0} \frac{\cos^{-1}(1 - h^2) \sin^{-1}(1 - h)}{h(1 - h^2)}$

$= \lim\limits_{h \to 0} \frac{\cos^{-1}(1 - h^2)}{h} \left(\frac{\sin^{-1} 1}{1} \right)$

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Let $\cos^{-1}\left(1-h^2\right)=\theta \Rightarrow \cos\theta=1-h^2$

$$=\frac{\pi}{2}\lim_{\theta\to 0}\frac{\theta}{\sqrt{1-\cos\theta}}$$

$$=\frac{\pi}{2}\lim_{\theta\to 0}\frac{1}{\sqrt{\frac{1-\cos\theta}{\theta^2}}}$$

$$=\frac{\pi}{2}\frac{1}{\sqrt{1/2}}$$

$$R=\frac{\pi}{\sqrt{2}}$$

Now finding left hand limit

$$L=\lim_{x\to 0^-} f\left(x\right)$$

$$=\lim_{h\to 0} f\left(-h\right)$$

$$=\lim_{h\to 0}\frac{\cos^{-1}\left(1-\{-h\}^2\right)\sin^{-1}\left(1-\{-h\}\right)}{\{-h\}-\{-h\}^3}$$

$$=\lim_{h\to 0}\frac{\cos^{-1}\left(1-(-h+1)^2\right)\sin^{-1}\left(1-(-h+1)\right)}{(-h+1)-(-h+1)^3}$$

$$=\lim_{h\to 0}\frac{\cos^{-1}\left(-h^2+2h\right)\sin^{-1}h}{(1-h)\left(1-(1-h)^2\right)}$$

$$=\lim_{h\to 0}\left(\frac{\pi}{2}\right)\frac{\sin^{-1}h}{\left(1-(1-h)^2\right)}$$

$$\frac{\pi}{2}\lim_{x\to 0}\left(\frac{\sin^{-1}h}{-h^2+2h}\right)$$

$$=\frac{\pi}{2}\lim_{h\to 0}\left(\frac{\sin^{-1}h}{h}\right)\left(\frac{1}{-h+2}\right)$$

$$L=\frac{\pi}{4}$$

$$\frac{32}{\pi^2}\left(L^2+R^2\right)=\frac{32}{\pi^2}\left(\frac{\pi^2}{2}+\frac{\pi^2}{16}\right)$$

= 16 + 2

= 18

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4. FUNDAMENTAL THEOREMS ON LIMITS:

Let $\lim_{x\to a} f(x) = l$ & $\lim_{x\to a} g(x) = m$. If l & m exist finitely then :

(a) Sum rule : $\lim_{x\to a}\{f(x)+g(x)\} = l+m$

(b) Difference rule : $\lim_{x\to a}\{f(x)-g(x)\} = l-m$

(c) Product rule : $\lim_{x\to a} f(x)\cdot g(x) = l\cdot m$

(d) Quotient rule : $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{l}{m}$, provided $m \neq 0$

(e) Constant multiple rule : $\lim_{x\to a} kf(x) = k\lim_{x\to a} f(x)$; where k is constant.

(f) Power rule : If m and n are integers then $\lim_{x\to a}[f(x)]^{m/n} = l^{m/n}$ provided $l^{m/n}$ is a real number.

(g) $\lim_{x\to a} f[g(x)] = f\left(\lim_{x\to a} g(x)\right) = f(m)$; provided $f(x)$ is continuous at $x = m$.

For example : $\lim_{x\to a} \ell n(g(x)) = \ell n[\lim_{x\to a} g(x)]$

$= \ell n(m)$; provided ℓnx is continuous at $x = m, m = \lim_{x\to a} g(x)$.

5. INDETERMINATE FORMS:

$\frac{0}{0}, \frac{\infty}{\infty}, \infty-\infty, 0\times\infty, 1^{\infty}, 0^{0}, \infty^{0}$

Initially we will deal with first five forms only and the other two forms will come up after we have gone through differentiation.

Note :

(i) Here 0,1 are not exact, in fact both are approaching to their corresponding values.

(ii) We cannot plot ∞ on the paper. Infinity (∞) is a symbol & not a number It does not obey the laws of elementary algebra,

(a) $\infty+\infty\to\infty$ (b) $\infty\times\infty\to\infty$ (c) $\infty^{\infty}\to\infty$ (d) $0^{\infty}\to 0$

6. GENERAL METHODS TO BE USED TO EVALUATE LIMITS:

(a) Factorization:

Important factors :

(i) $x^n - a^n = (x-a)(x^{n-1} + ax^{n-2} + \ldots\ldots\ldots + a^{n-1}), n \in N$

(ii) $x^n + a^n = (x+a)(x^{n-1} - ax^{n-2} + \ldots\ldots\ldots + a^{n-1})$, n is an odd natural number.

Note : $\lim_{x\to a}\frac{x^n - a^n}{x-a} = na^{n-1}$

Illustration 6

Evaluate: $\lim_{x\to 2}\left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3-3x^2+2x}\right]$

Solution:

We have

$$\lim_{x\to 2}\left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3-3x^2+2x}\right] = \lim_{x\to 2}\left[\frac{1}{x-2} - \frac{2(2x-3)}{x(x-1)(x-2)}\right] = \lim_{x\to 2}\left[\frac{x(x-1)-2(2x-3)}{x(x-1)(x-2)}\right]$$

$$= \lim_{x\to 2}\left[\frac{x^2-5x+6}{x(x-1)(x-2)}\right] = \lim_{x\to 2}\left[\frac{(x-2)(x-3)}{x(x-1)(x-2)}\right] = \lim_{x\to 2}\left[\frac{x-3}{x(x-1)}\right] = -\frac{1}{2}$$

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(b) Rationalization or Double Rationalization:

Illustration 7

Evaluate: $\lim\limits_{x\to1}\dfrac{4-\sqrt{15x+1}}{2-\sqrt{3x+1}}$

Solution:

$$\lim_{x\to1}\frac{4-\sqrt{15x+1}}{2-\sqrt{3x+1}}=\lim_{x\to1}\frac{(4-\sqrt{15x+1})(2+\sqrt{3x+1})(4+\sqrt{15x+1})}{(2-\sqrt{3x+1})(4+\sqrt{15x+1})(2+\sqrt{3x+1})}$$

$$=\lim_{x\to1}\frac{(15-15x)}{(3-3x)}\times\frac{2+\sqrt{3x+1}}{4+\sqrt{15x+1}}=\frac{5}{2}$$

Illustration 8

Evaluate: $\lim\limits_{x\to1}\left(\dfrac{\sqrt{x^2+8}-\sqrt{10-x^2}}{\sqrt{x^2+3}-\sqrt{5-x^2}}\right)$

Solution:

This is of the form $\dfrac{3-3}{2-2}=\dfrac{0}{0}$ if we put $x=1$

To eliminate the $\dfrac{0}{0}$ factor, multiply by the conjugate of numerator and the conjugate of the denominator

$$\therefore \text{ Limit } = \lim_{x\to1}\left(\sqrt{x^2+8}-\sqrt{10-x^2}\right)\frac{(\sqrt{x^2+8}+\sqrt{10-x^2})}{(\sqrt{x^2+8}+\sqrt{10-x^2})}\times\frac{(\sqrt{x^2+3}+\sqrt{5-x^2})}{(\sqrt{x^2+3}+\sqrt{5-x^2})(\sqrt{x^2+3}-\sqrt{5-x^2})}$$

$$=\lim_{x\to1}\frac{\sqrt{x^2+3}+\sqrt{5-x^2}}{\sqrt{x^2+8}+\sqrt{10-x^2}}\times\frac{(x^2+8)-(10-x^2)}{(x^2+3)-(5-x^2)}=\lim_{x\to1}\left(\frac{\sqrt{x^2+3}+\sqrt{5-x^2}}{\sqrt{x^2+8}+\sqrt{10-x^2}}\right)\times1=\frac{2+2}{3+3}=\frac{2}{3}$$

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Illustration 9

If $a=\lim\limits_{x\to0}\dfrac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$ and $b=\lim\limits_{x\to0}\dfrac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$, then the value of ab^3 is :

(A) 36 (B) 32 (C) 25 (D) 30

Ans. (B)

Solution:

$$a=\lim_{x\to0}\frac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$$

$$=\lim_{x\to0}\frac{\sqrt{1+x^4}-1}{x^4\left(\sqrt{1+\sqrt{1+x^4}}+\sqrt{2}\right)}$$

$$=\lim_{x\to0}\frac{x^4}{x^4\left(\sqrt{1+\sqrt{1+x^4}}+\sqrt{2}\right)\left(\sqrt{1+x^4}+1\right)}$$

Applying limit $a=\dfrac{1}{4\sqrt{2}}$

$$b=\lim_{x\to0}\frac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$$

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$$=\lim_{x\to 0}\frac{\left(1-\cos^2 x\right)\left(\sqrt{2}+\sqrt{1+\cos x}\right)}{2-\left(1+\cos x\right)}$$

$$b=\lim_{x\to 0}\left(1+\cos x\right)\left(\sqrt{2}+\sqrt{1+\cos x}\right)$$

Applying limits $b=2\left(\sqrt{2}+\sqrt{2}\right)=4\sqrt{2}$

Now, $ab^3=\frac{1}{4\sqrt{2}}\times\left(4\sqrt{2}\right)^3=32$

(c) Limit When $x \to \infty$:

(i) Divide by greatest power of x in numerator and denominator.

(ii) Put $x = 1/y$ and apply $y \to 0$

Illustration 10

Evaluate: $\lim_{x\to\infty}\frac{x^2+x+1}{3x^2+2x-5}$

Solution:

$\lim_{x\to\infty}\frac{x^2+x+1}{3x^2+2x-5}$, $\left(\frac{\infty}{\infty}\text{ form}\right)$

Put $x = \frac{1}{y}$

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Limit = $\lim_{y\to 0}\frac{1+y+y^2}{3+2y-5y^2}=\frac{1}{3}$

Illustration 11

If $\lim_{x\to\infty}\left(\frac{x^3+1}{x^2+1}-(ax+b)\right)=2$, then

(A) $a = 1, b = 1$ (B) $a = 1, b = 2$ (C) $a = 1, b = -2$ (D) none of these

Ans. (C)

Solution:

$$\lim_{x\to\infty}\left(\frac{x^3+1}{x^2+1}-(ax+b)\right)=2\Rightarrow\lim_{x\to\infty}\frac{x^3(1-a)-bx^2-ax+(1-b)}{x^2+1}=2$$

$$\Rightarrow\lim_{x\to\infty}\frac{x(1-a)-b-\frac{a}{x}+\frac{(1-b)}{x^2}}{1+\frac{1}{x^2}}=2\Rightarrow 1-a=0,-b=2\Rightarrow a=1,b=-2$$

Illustration 12

If $\lim_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$ then $8(\alpha+\beta)$ is equal to :

(A) 4 (B) –8 (C) –4 (D) 8

Ans. (C)

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Solution:

$$\lim_{n\to\infty} n\left(1-\frac{n+1}{n^2}\right)^{\frac{1}{2}}+\alpha n+\beta=0$$

$$\lim_{n\to\infty} n\left\{1-\frac{1}{2}\left(\frac{n+1}{n^2}\right)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}\left(\frac{n+1}{n^2}\right)^2+....\right\}+\alpha n+\beta=0$$

$$\lim_{n\to\infty} n-\frac{1}{2}+\frac{1}{n}+....+n\alpha+\beta=0$$

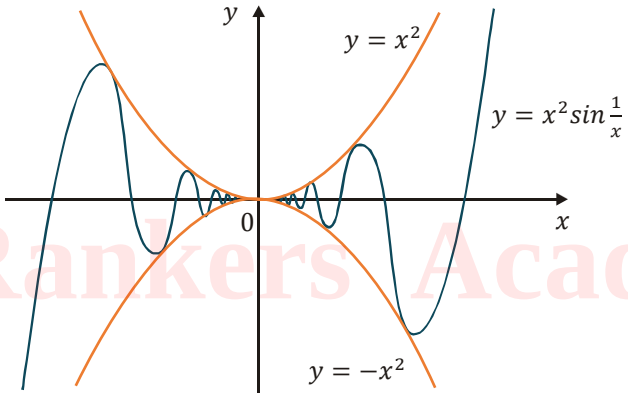
$$\alpha=-1,\beta=\frac{1}{2}$$

$$8(\alpha+\beta)=-4$$

(d) Squeeze Play Theorem (Sandwich Theorem) :

Statement : If $f(x)\le g(x)\le h(x)$; $\forall\ x$ in the neighbourhood at $x=a$ and $\lim\limits_{x\to a} f(x) = \ell = \lim\limits_{x\to a} h(x)$ then, $\lim\limits_{x\to a} g(x)=\ell$

Ex.1 $\lim\limits_{x\to 0} x^2\sin\frac{1}{x}=0$,



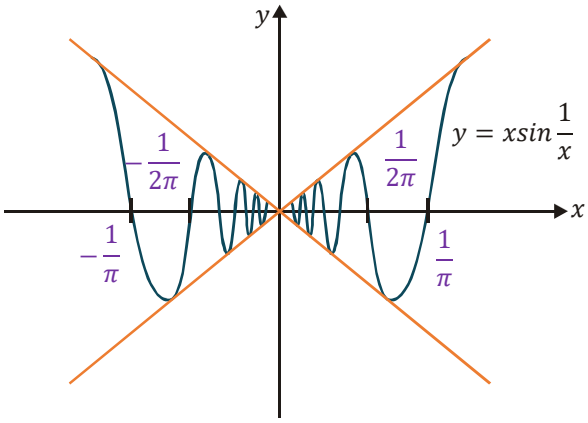
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$\because$ $\sin\left(\frac{1}{x}\right)$ lies between –1 & 1

$\Rightarrow -x^2\le x^2\sin\frac{1}{x}\le x^2$

$\Rightarrow \lim\limits_{x\to 0} x^2\sin\frac{1}{x}=0$ as $\lim\limits_{x\to 0}(-x^2)=\lim\limits_{x\to 0} x^2=0$

Ex.2 $\lim\limits_{x\to 0} x\sin\frac{1}{x}=0$



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$\because \quad \sin\left(\frac{1}{x}\right)$ lies between –1 & 1

$\Rightarrow -x \le x\sin\frac{1}{x} \le x$

$\Rightarrow \lim\limits_{x\to 0} x\sin\frac{1}{x} = 0$ as $\lim\limits_{x\to 0}(-x) = \lim\limits_{x\to 0} x = 0$

Illustration 13

Evaluate: $\lim\limits_{n\to\infty}\frac{[x]+[2x]+[3x]+.....[nx]}{n^2}$ (Where [.] denotes the greatest integer function.)

Solution:

We know that $x - 1 < [x] \le x$

$\Rightarrow x + 2x + \ldots . nx - n < \sum\limits_{r=1}^{n}[rx] \le x + 2x + \ldots\ldots + nx$

$\Rightarrow \frac{xn}{2}(n+1) - n < \sum\limits_{r=1}^{n}[rx] \le \frac{x.n(n+1)}{2}$

$\Rightarrow \frac{x}{2}\left(1+\frac{1}{n}\right) - \frac{1}{n} < \frac{1}{n^2}\sum\limits_{r=1}^{n}[rx] \le \frac{x}{2}\left(1+\frac{1}{n}\right)$

Now, $\lim\limits_{n\to\infty}\frac{x}{2}\left(1+\frac{1}{n}\right) = \frac{x}{2}$ and $\lim\limits_{n\to\infty}\frac{x}{2}\left(1+\frac{1}{n}\right) - \frac{1}{n} = \frac{x}{2}$

Thus, $\lim\limits_{n\to\infty}\frac{[x]+[2x]+.....+[nx]}{n^2} = \frac{x}{2}$

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Illustration 14

$\lim\limits_{n\to\infty}\left(1+\frac{1+\frac{1}{2}+.....+\frac{1}{n}}{n^2}\right)^n$ is equal to :

(A) $\frac{1}{2}$ (B) 0 (C) $\frac{1}{e}$ (D) 1

Ans. (D)

Solution:

Given limit is of 1^∞ form

So, $l = \exp\left(\lim\limits_{n\to\infty}\frac{1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n}}{n}\right)$

Now,

$0 \le 1+\frac{1}{2}+\frac{1}{3}+....+\frac{1}{n} \le 1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+....+\frac{1}{\sqrt{n}}$

So, l = exp (0) (from sandwich theorem)

= 1

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Illustration 15

Let $f:\to R\to(0,\infty)$ be strictly increasing function such that $\lim\limits_{x\to\infty}\dfrac{f(7x)}{f(x)}=1$. Then, the value of $\lim\limits_{x\to\infty}\left[\dfrac{f(5x)}{f(x)}-1\right]$ is equal to

(A) 4 (B) 0 (C) 7/5 (D) 1

Ans. (B)

Solution:

$f:R\to(0,\infty)$

$\lim\limits_{x\to\infty}\dfrac{f(7x)}{f(x)}=1$

$\because f$ is increasing

$\therefore f(x)<f(5x)<f(7x)$

$\because \dfrac{f(x)}{f(x)}<\dfrac{f(5x)}{f(x)}<\dfrac{f(7x)}{f(x)}$

$1<\lim\limits_{x\to\infty}\dfrac{f(5x)}{f(x)}<1$

$\therefore\left[\dfrac{f(5x)}{f(x)}-1\right]$

$\Rightarrow 1-1=0$

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7. LIMIT OF TRIGONOMETRIC FUNCTIONS:

$\lim\limits_{x\to0}\dfrac{\sin x}{x}=1=\lim\limits_{x\to0}\dfrac{\tan x}{x}=\lim\limits_{x\to0}\dfrac{\tan^{-1}x}{x}=\lim\limits_{x\to0}\dfrac{\sin^{-1}x}{x}$ [where x is measured in radians]

If $\lim\limits_{x\to a}f(x)=0$, then $\lim\limits_{x\to a}\dfrac{\sin f(x)}{f(x)}=1$, e.g., $\lim\limits_{x\to1}\dfrac{\sin(\ell nx)}{(\ell nx)}=1$

Illustration 16

Evaluate: $\lim\limits_{x\to0}\dfrac{x^3\cot x}{1-\cos x}$

Solution:

$\lim\limits_{x\to0}\dfrac{x^3\cos x}{\sin x(1-\cos x)}=\lim\limits_{x\to0}\dfrac{x^3\cos x(1+\cos x)}{\sin x.\sin^2 x}=\lim\limits_{x\to0}\dfrac{x^3}{\sin^3 x}.\cos x(1+\cos x)=2$

Illustration 17

Evaluate: $\lim\limits_{x\to0}\dfrac{(2+x)\sin(2+x)-2\sin 2}{x}$

Solution:

$\lim\limits_{x\to0}\dfrac{2(\sin(2+x)-\sin 2)+x\sin(2+x)}{x}=\lim\limits_{x\to0}\left(\dfrac{2.2.\cos\left(2+\dfrac{x}{2}\right)\sin\dfrac{x}{2}}{x}+\sin(2+x)\right)$

$=\lim\limits_{x\to0}\dfrac{2\cos\left(2+\dfrac{x}{2}\right)\sin\dfrac{x}{2}}{\dfrac{x}{2}}+\lim\limits_{x\to0}\sin(2+x)=2\cos2+\sin 2$

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Illustration 18

Evaluate: $\lim\limits_{n\to\infty}\dfrac{\sin\dfrac{a}{n}}{\tan\dfrac{b}{n+1}}$

Solution:

As $n\to\infty$, $\dfrac{1}{n}\to 0$ and $\dfrac{a}{n}$ also tends to zero

$\sin\dfrac{a}{n}$ should be written as $\dfrac{\sin\dfrac{a}{n}}{\dfrac{a}{n}}$ so that it looks like $\lim\limits_{\theta\to 0}\dfrac{\sin\theta}{\theta}$

The given limit $=\lim\limits_{n\to\infty}\left(\dfrac{\sin\dfrac{a}{n}}{\dfrac{a}{n}}\right)\left(\dfrac{\dfrac{b}{n+1}}{\tan\dfrac{b}{n+1}}\right).\dfrac{a(n+1)}{n.b}$

$=\lim\limits_{n\to\infty}\left(\dfrac{\sin\dfrac{a}{n}}{\dfrac{a}{n}}\right)\left(\dfrac{\dfrac{b}{n+1}}{\tan\dfrac{b}{n+1}}\right).\dfrac{a}{b}\left(1+\dfrac{1}{n}\right)=1\times 1\times\dfrac{a}{b}\times 1=\dfrac{a}{b}$

Illustration 19

$\lim\limits_{x\to 0}\dfrac{x\cot(4x)}{\sin^2 x\cot^2(2x)}$ is equal to :-

(A) 2 (B) 0 (C) 4 (D) 1

Ans. (D)

Solution:

$$\lim_{x\to 0}\frac{x\tan^2 2x}{\tan 4x\sin^2 x}=\lim_{x\to 0}\frac{x\left(\dfrac{\tan^2 2x}{4x^2}\right)4x^2}{\left(\dfrac{\tan 4x}{4x}\right)4x\left(\dfrac{\sin^2 x}{x^2}\right)x^2}=1$$

Illustration 20

The value of $\lim\limits_{x\to 1}\dfrac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ is equal to:

(A) $\dfrac{\pi^2}{6}$ (B) $\dfrac{\pi^2}{3}$ (C) $\dfrac{\pi^2}{2}$ (D) π^2

Ans. (D)

Solution:

$$\lim_{x\to 1}\frac{(x^2-1)\sin^2\pi x}{(x^2-1)(x-1)^2}=\lim_{x\to 1}\left(\frac{\sin((1-x)\pi)}{\pi(1-x)}\right)^2\pi^2=\pi^2$$

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Illustration 21

The value of the limit $\lim\limits_{\theta\to 0}\dfrac{\tan\left(\pi\cos^2\theta\right)}{\sin\left(2\pi\sin^2\theta\right)}$ is equal to:

(A) $\lim\limits_{\theta\to 0}\dfrac{\tan\left(\pi\cos^2\theta\right)}{\sin\left(2\pi\sin^2\theta\right)}$ (B) $-\dfrac{1}{4}$ (C) 0 (D) $\dfrac{1}{4}$

Ans. (A)

Solution:

$$\lim_{\theta\to 0}\frac{\tan\left(\pi(1-\sin^2\theta)\right)}{\sin\left(2\pi\sin^2\theta\right)}$$

$$=\lim_{\theta\to 0}\frac{-\tan\left(\pi\sin^2\theta\right)}{\sin\left(2\pi\sin^2\theta\right)}$$

$$=\lim_{\theta\to 0}-\left(\frac{\tan\left(\pi\sin^2\theta\right)}{\pi\sin^2\theta}\right)\left(\frac{2\pi\sin^2\theta}{\sin\left(2\pi\sin^2\theta\right)}\right)\times\frac{1}{2}$$

$$=\frac{-1}{2}$$

Illustration 22

If $\lim\limits_{x\to 0}\left\{\dfrac{1}{x^8}\left(1-\cos\dfrac{x^2}{2}-\cos\dfrac{x^2}{4}+\cos\dfrac{x^2}{2}\cos\dfrac{x^2}{4}\right)\right\}=2^{-k}$, then the value of k is _____

Ans. (8)

Solution:

$$\lim_{x\to 0}\left\{\frac{1}{x^8}\left(1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}\cos\frac{x^2}{4}\right)\right\}=2^{-k}$$

$$\Rightarrow\lim_{x\to 0}\frac{\left(1-\cos\dfrac{x^2}{2}\right)}{4\left(\dfrac{x^2}{2}\right)^2}\frac{\left(1-\cos\dfrac{x^2}{4}\right)}{16\left(\dfrac{x^2}{4}\right)^2}=\frac{1}{8}\times\frac{1}{32}=2^{-k}$$

$\Rightarrow 2^{-8}=2^{-k}\Rightarrow k=8.$

Illustration 23

If α is the positive root of the equation, $p(x)=x^2-x-2=0$, then $\lim\limits_{x\to\alpha^+}\dfrac{\sqrt{1-\cos\left(p(x)\right)}}{x+\alpha-4}$ is equal to

(A) $\dfrac{3}{\sqrt{2}}$ (B) $\dfrac{3}{2}$ (C) $\dfrac{1}{\sqrt{2}}$ (D) $\dfrac{1}{2}$

Ans. (A)

Solution:

$x^2-x-2=0$

roots are 2 & –1

$$\Rightarrow\lim_{x\to 2^+}\frac{\sqrt{1-\cos(x^2-x-2)}}{(x-2)}$$

$$=\lim_{x\to 2^+}\frac{\sqrt{2\sin^2\dfrac{(x^2-x-2)}{2}}}{(x-2)}$$

$$=\lim_{x\to 2^+}\frac{\sqrt{2}\sin\left(\dfrac{(x-2)(x+1)}{2}\right)}{(x-2)}=\frac{3}{\sqrt{2}}$$

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Illustration 24

If $\alpha > \beta > 0$ are the roots of the equation $ax^2 + bx + 1 = 0$, and $\lim\limits_{x\to\frac{1}{\alpha}}\left(\frac{1-\cos\left(x^2+bx+a\right)}{2\left(1-\alpha x\right)^2}\right)^{\frac{1}{2}} = \frac{1}{k}\left(\frac{1}{\beta}-\frac{1}{\alpha}\right)$, then k is equal to

(A) 2β (B) 2α (C) α (D) β

Ans. (B)

Solution:

α, β are roots of $ax^2 + bx + 1 = 0$

$\frac{1}{\alpha}, \frac{1}{\beta}$ are roots of $x^2 + bx + a = 0$,

(by transformation)

$$x^2 + bx + a = \left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)$$

$$\lim_{x\to\frac{1}{\alpha}}\left[\frac{1-\cos\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)}{2\left(1-\alpha x\right)^2}\right]^{\frac{1}{2}} = L$$

$$\left(\text{By using } \lim_{\theta\to 0}\frac{1-\cos\theta}{\theta^2} = \frac{1}{2}\right)$$

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$$\Rightarrow \left[\frac{\left(\frac{1}{\alpha}-\frac{1}{\beta}\right)^2}{4\alpha^2}\right]^{\frac{1}{2}} = L$$

$$\Rightarrow \frac{\frac{1}{\beta}-\frac{1}{\alpha}}{2\alpha} = L$$

Comparing $k = 2\alpha$

8. LIMIT OF EXPONENTIAL FUNCTIONS:

(a) $\lim\limits_{x\to 0}\frac{a^x-1}{x} = \ell n a (a>0)$ In particular $\lim\limits_{x\to 0}\frac{e^x-1}{x} = 1$.

In general, if $\lim\limits_{x\to a} f(x) = 0$, then $\lim\limits_{x\to a}\frac{a^{f(x)}-1}{f(x)} = \ell n a,\ a>0$

Illustration 25

Evaluate: $\lim\limits_{x\to 0}\frac{e^{\tan x}-e^x}{\tan x - x}$

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Solution:

$$\lim_{x\to0}\frac{e^{\tan x}-e^{x}}{\tan x-x}=\lim_{x\to0}\frac{e^{x}\times e^{(\tan x-x)}-e^{x}}{\tan x-x}$$

$$=\lim_{x\to0}\frac{e^{x}(e^{\tan x-x}-1)}{\tan x-x}=\lim_{x\to0\,y\to0}\frac{e^{x}(e^{y}-1)}{y}\text{ where } y=\tan x-x \text{ and } \lim_{y\to0}\frac{e^{y}-1}{y}=1$$

$= e^{0}\times 1$ [as $x\to 0, \tan x - x\to 0$]

$= 1 \times 1 = 1$

(b) (i) $\lim_{x\to0}(1+x)^{1/x}=e=\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}$ (**Note:** The base and exponent depend on the same variable.)

In general, if $\lim_{x\to a} f(x)=0$, then $\lim_{x\to a}(1+f(x))^{1/f(x)}=e$

(ii) $\lim_{x\to0}\frac{\ell n(1+x)}{x}=1$

(iii) If $\lim_{x\to a} f(x)=1$ and $\lim_{x\to a}\phi(x)=\infty$, then; $\lim_{x\to a}\left[f(x)\right]^{\phi(x)}=e^{k}$

where $k=\lim_{x\to a}\phi(x)\,[f(x)-1]$

Illustration 26

$$\lim_{x\to0}\frac{x\left(e^{\left(\sqrt{1+x^{2}+x^{4}}-1\right)/x}-1\right)}{\sqrt{1+x^{2}+x^{4}}-1}$$

(A) does not exist. (B) is equal to $\sqrt{e}$. (C) is equal to 0. (D) is equal to 1.

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Ans. (D)

Solution:

$$\lim_{x\to0}\frac{x\left(e^{\left(\sqrt{1+x^{2}+x^{4}}-1\right)/x}-1\right)}{\sqrt{1+x^{2}+x^{4}}-1}$$

$\because \lim_{x\to0}\frac{\sqrt{1+x^{2}+x^{4}}-1}{x}$ (from)

$$\lim_{x\to0}\frac{(1+x^{2}+x^{4})-1}{x(\sqrt{1+x^{2}+x^{4}}+1}$$

$$\lim_{x\to0}\frac{x(1+x^{2})}{\left(\sqrt{1+x^{2}+x^{4}}+1\right)}=0$$

So $\lim_{x\to0}\frac{x\left(e^{\left(\frac{\sqrt{1+x^{2}+x^{4}}-1}{x}\right)}-1\right)}{\sqrt{1+x^{2}+x^{4}}-1}$ ($\frac{0}{0}$ from)

$$\lim_{x\to0}\frac{e^{\frac{\sqrt{1+x^{2}+x^{4}}-1}{x}}-1}{\left(\frac{\sqrt{1+x^{2}+x^{4}}-1}{x}\right)}=1$$

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Illustration 27

Evaluate: $\lim_{x\to 1}(\log_3 3x)^{\log_x 3}$

Solution:

$\lim_{x\to 1}(\log_3 3x)^{\log_x 3} = \lim_{x\to 1}(\log_3 3+\log_3 x)^{\log_x 3}$

$=\lim_{x\to 1}(1+\log_3 x)^{1/\log_3 x} = e \qquad \because \log_b a=\frac{1}{\log_a b}$

Illustration 28

Evaluate: $\lim_{x\to 0}\frac{x\ell n(1+2\tan x)}{1-\cos x}$

Solution:

$\lim_{x\to 0}\frac{x\ell n(1+2\tan x)}{1-\cos x}=\lim_{x\to 0}\frac{x\ell n(1+2\tan x)}{\frac{1-\cos x}{x^2}.x^2}.\frac{2\tan x}{2\tan x}=4$

Illustration 29

$\lim_{x\to 0}\frac{\log_e\left(1-x+x^2\right)+\log_e\left(1+x+x^2\right)}{\sec x-\cos x}$ is equal to :

(A) 1 (B) –1 (C) e (D) 0

Ans. (A)

Solution:

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$\lim_{x\to 0}\frac{\left(\ln\left(1+x^2+x^4\right)\right)\cos x}{1-\cos^2 x}$

$\lim_{x\to 0}\frac{\left(\frac{\ln\left(1+x^2+x^4\right)}{x^2+x^4}\right)x^2\left(1+x^2\right)\cos x}{\left(\frac{\sin^2 x}{x^2}\right)x^2}=1$

Illustration 30

Evaluate: $\lim_{x\to\infty}\left(\frac{2x^2-1}{2x^2+3}\right)^{4x^2+2}$

Solution:

Since it is in the form of 1^∞

$\lim_{x\to\infty}\left(\frac{2x^2-1}{2x^2+3}\right)^{4x^2+2}=e^{\lim_{x\to\infty}\left(\frac{2x^2-1-2x^2-3}{2x^2+3}\right)(4x^2+2)}=e^{-8}$

Illustration 31

Evaluate : $\lim_{x\to\infty}\left(\frac{7x^2+1}{5x^2-1}\right)^{\frac{x^5}{1-x^3}}$

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Solution:

Here $f(x)=\frac{7x^2+1}{5x^2-1}$, $\phi(x)=\frac{x^5}{1-x^3}=\frac{x^2.x^3}{1-x^3}=\frac{x^2}{\frac{1}{x^3}-1}$

$\therefore \lim\limits_{x\to\infty} f(x)=\frac{7}{5}$ & $\lim\limits_{x\to\infty}\phi(x)\to-\infty$

$\Rightarrow \lim\limits_{x\to\infty}(f(x))^{\phi(x)}=\left(\frac{7}{5}\right)^{-\infty}=0$

Illustration 32

$\lim\limits_{x\to 0}\left(\frac{1+\tan x}{1+\sin x}\right)^{\operatorname{cosec} x}$ is equal to

(A) e (B) $\frac{1}{e}$ (C) 1 (D) None of these

Ans. (C)

Solution:

Given limit = $\lim\limits_{x\to 0}[(1+\tan x)^{\operatorname{cosec} x}\times 1/(1+\sin x)^{\operatorname{cosec} x}]$

$=\lim\limits_{x\to 0}[\{(1+\tan x)^{\cot x}\}^{\sec x}\times\{1/(1+\sin x)^{cosec\, x}\}]=e^{\sec 0}.\frac{1}{e}=e.\frac{1}{e}=1$

Illustration 33

If $\lim\limits_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x}=e^2$ then the values of a and b are

(A) $a=1, b=2$ (B) $a=1, b\in R$ (C) $a\in R, b=2$ (D) $a\in R, b\in R$

Ans. (B)

Solution:

Since, $\lim\limits_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)=1$

$\therefore \lim\limits_{x\to\infty}\left[\left(1+\frac{ax+b}{x^2}\right)^{\frac{x^2}{ax+b}}\right]^{\frac{2(ax+b)}{x}}=e^2\Rightarrow\lim\limits_{x\to\infty}\frac{2(ax+b)}{x}=e^2\Rightarrow\lim\limits_{x\to\infty}\frac{2(ax+b)}{x}=2\Rightarrow 2a=2\Rightarrow a=1$

Thus $a=1$ and $b\in R$.

Illustration 34

$Lim\limits_{x\to 0}\frac{e-(1+2x)^{\frac{1}{2x}}}{x}$ is equal to :

(A) e (B) $\frac{-2}{e}$ (C) 0 (D) $e-e^2$

Ans. (A)

Solution:

$Lim\limits_{x\to 0}\frac{e-e^{\frac{1}{2x}\ln(1+2x)}}{x}=Lim\limits_{x\to 0}(-e)\frac{\left(e^{\frac{\ln(1+2x)}{2x}-1}-1\right)}{x}$

$=Lim\limits_{x\to 0}(-e)\frac{\ln(1+2x)-2x}{2x^2}$

$=(-e)\times(-1)\frac{4}{2\times 2}=e$

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Illustration 35

Given $h(x)=\lim\limits_{n\to\infty}\dfrac{x^{2n}f(x)+g(x)}{x^{2n}+1}$, $f(2)=5$ and $g\left(\dfrac{1}{2}\right)=-3$, then value of $h(2)-2h\left(\dfrac{1}{2}\right)$ is ..... .
(Given $f(x)$ and $g(x)$ are bounded functions)

Ans. (11)

Solution:

$$h(x)=\lim_{n\to\infty}\frac{x^{2n}f(x)+g(x)}{x^{2n}+1}=\begin{cases}g(x) & |x|<1\\ \dfrac{f(x)+g(x)}{2} & |x|=1\\ f(x) & |x|>1\end{cases}$$

$$\therefore h(2)-2h\left(\frac{1}{2}\right)=f(2)-2g\left(\frac{1}{2}\right)=5-2(-3)=11$$

9. LIMIT USING SERIES EXPANSION:

Expansion of function like binomial expansion, exponential & logarithmic expansion, expansion of $\sin x, \cos x, \tan x$ should be remembered by heart which are given below :

(a) $a^x=1+\dfrac{x\,\ell na}{1!}+\dfrac{x^2\ell n^2a}{2!}+\dfrac{x^3\ell n^3a}{3!}+\ldots, x\in\mathbb{R}, a>0, a\neq 1$

(b) $e^x=1+\dfrac{x}{1!}+\dfrac{x^2}{2!}+\dfrac{x^3}{3!}+\ldots, x\in\mathbb{R}$

(c) $\ell n(1+x)=x-\dfrac{x^2}{2}+\dfrac{x^3}{3}-\dfrac{x^4}{4}+\ldots$ for $-1<x\le 1$

(d) $\sin x=x-\dfrac{x^3}{3!}+\dfrac{x^5}{5!}-\dfrac{x^7}{7!}+\ldots, x\in\mathbb{R}$

(e) $\cos x=1-\dfrac{x^2}{2!}+\dfrac{x^4}{4!}-\dfrac{x^6}{6!}+\ldots, x\in\mathbb{R}$

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(f) $\tan x=x+\dfrac{x^3}{3}+\dfrac{2x^5}{15}+\ldots, -\dfrac{\pi}{2}<x<\dfrac{\pi}{2}$

(g) $\tan^{-1}x=x-\dfrac{x^3}{3}+\dfrac{x^5}{5}-\dfrac{x^7}{7}+\ldots, x\in(-1,1)$

(h) $\sin^{-1}x=x+\dfrac{1^2}{3!}x^3+\dfrac{1^2.3^2}{5!}x^5+\dfrac{1^2.3^2.5^2}{7!}x^7+\ldots, x\in(-1,1)$

(i) $\sec^{-1}x=1+\dfrac{x^2}{2!}+\dfrac{5x^4}{4!}+\dfrac{61x^6}{6!}+\ldots, x\in(-\infty,-1)\cup(1,\infty)$

(j) $(1+x)^n=1+nx+\dfrac{n(n-1)}{2!}x^2+\ldots, n\in\mathbb{R}, x\in(-1,1)$

Illustration 36

$\lim\limits_{x\to 0}\dfrac{e^x-e^{-x}-2x}{x-\sin x}$

Solution:

$$\lim_{x\to0}\frac{e^x-e^{-x}-2x}{x-\sin x}\Rightarrow\lim_{x\to0}\frac{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots..-\left(1-x+\frac{x^2}{2!}-\frac{x^3}{3!}+\ldots..\right)-2x}{x-\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}.....\right)}$$

$$\Rightarrow\lim_{x\to0}\frac{2.\frac{x^3}{6}+2.\frac{x^5}{5!}+\ldots...}{\frac{x^3}{6}+\frac{x^5}{5!}.....}\Rightarrow\lim_{x\to0}\frac{x^3\left(\frac{1}{3}+\frac{1}{60}x^2+\ldots..\right)}{x^3\left(\frac{1}{6}+\frac{1}{120}x^2+\ldots..\right)}=\frac{1/3}{1/6}=2$$

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Illustration 37

If $\lim_{x\to 0}\frac{\sin^{-1}x-\tan^{-1}x}{3x^3}$ is equal to L, then the value of $(6L+1)$ is

(A) $\frac{1}{6}$ (B) $\frac{1}{2}$ (C) 6 (D) 2

Ans. (D)

Solution:

$$\lim_{x\to 0}\frac{\left(x+\frac{x^3}{3!}\ldots\right)-\left(x-\frac{x^3}{3}\ldots\right)}{3x^3}=\frac{1}{6}$$

So $6L+1=2$

Illustration 38

If $\lim_{x\to 0}\frac{ax^2e^x-b\log_e(1+x)+cxe^{-x}}{x^2\sin x}=1$, then $16(a^2+b^2+c^2)$ is equal to _____.

Ans. (81)

Solution:

$$\lim_{x\to 0}\frac{ax^2\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+.....\right)-b\left(x-\frac{x^2}{2}+\frac{x^3}{3}-.....\right)+cx\left(1-x+\frac{x^2}{x!}-\frac{x^3}{3!}+.....\right)}{x^3.\frac{\sin x}{x}}$$

$$=\lim_{x\to\infty}\frac{(c-b)x+\left(\frac{b}{2}-c+a\right)x^2+\left(a-\frac{b}{3}+\frac{c}{2}\right)x^3+.....}{x^3}=1$$

$$c-b=0,\quad \frac{b}{2}-c+a=0$$

$$a-\frac{b}{3}+\frac{c}{2}=1\quad a=\frac{3}{4}\quad b=c=\frac{3}{2}$$

$$a^2+b^2+c^2=\frac{9}{16}+\frac{9}{4}+\frac{9}{4}$$

$$16\left(a^2+b^2+c^2\right)=81$$

Illustration 39

If $\lim_{x\to 0}\frac{3+\alpha\sin x+\beta\cos x+\log_e(1-x)}{3\tan^2 x}=\frac{1}{3}$, then $2\alpha-\beta$ is equal to:

(A) 2 (B) 7 (C) 5 (D) 1

Ans. (C)

Solution:

$$\lim_{x\to 0}\frac{3+\alpha\sin x+\beta\cos x+\log_e(1-x)}{3\tan^2 x}=\frac{1}{3}$$

$$\Rightarrow\lim_{x\to 0}\frac{3+\alpha\left[x-\frac{x^3}{3!}+....\right]+\beta\left[1-\frac{x^2}{2!}+\frac{x^4}{4!}....\right]+\left(-x-\frac{x^2}{2}-\frac{x^3}{3}\ldots\right)}{3\tan^2 x}=\frac{1}{3}$$

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$$\Rightarrow \lim_{x\to 0}\frac{(3+\beta)+(\alpha-1)x+\left(-\frac{1}{2}-\frac{\beta}{2}\right)x^2+....}{3x^2}\times\frac{x^2}{\tan^2 x}=\frac{1}{3}$$

$$\Rightarrow \beta+3=0,\ \alpha-1=0 \text{ and } \frac{-\frac{1}{2}-\frac{\beta}{2}}{3}=\frac{1}{3}$$

$$\Rightarrow \beta=-3, \alpha=1$$

$$\Rightarrow 2\alpha-\beta=2+3=5$$

Illustration 40

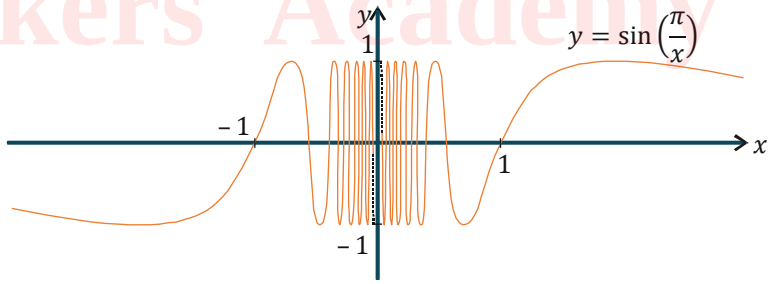
Evaluate $\lim_{x\to 0} \sin\frac{\pi}{x}$.

Solution:

Again, the function $f(x) = \sin(\pi/x)$ is undefined at 0. Evaluating the function for some small values of x, we get $f(1) = \sin\pi = 0$, $f\left(\frac{1}{2}\right)=\sin 2\pi=0$,

$f(0.1) = \sin 10\pi = 0, f(0.01) = \sin 100\pi = 0.$

Based on this information, we might be tempted to guess that $\lim_{x\to 0}\sin\frac{\pi}{x}=0$ but this time our guess is wrong. Note that although $f(1/n) = \sin n\pi = 0$ for any integer n, it is also true that $f(x) = 1$ for infinitely many values of x that approach 0. [In fact, $\sin(\pi/x) = 1$ when $\frac{\pi}{x}=\frac{\pi}{2}+2n\pi$ and solving for x, we get $x = 2/(4n + 1)$]. The graph of f is given in following figure



The dashed line indicates that the values of $\sin(\pi/x)$ oscillate between 1 and –1 infinitely often as x approaches 0. Since the values of $f(x)$ do not approach a fixed number as x approaches 0,

$\Rightarrow \lim_{x\to 0}\sin\frac{\pi}{x}$ does not exist.

Illustration 41

$$\lim_{x\to 0}\left\{\frac{\sin x - x + \frac{x^3}{6}}{x^5}\right\} =$$

(A) 1/120 (B) –1/120 (C) 1/20 (D) None of these

Ans. (A)

Solution:

Expand $\sin x$ and then solve.

$$\sin x = x - \frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\ldots\ldots \qquad \lim_{x\to 0}\frac{x-\frac{x^3}{6}+\frac{x^5}{120}-\ldots\ldots-x+\frac{x^3}{6}}{x^5}=\frac{1}{120}$$

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Illustration 42

Let $f:(-\infty,\infty)-\{0\}\to R$ be a differentiable function such that $f'(1)=\lim_{a\to\infty}a^2 f\left(\frac{1}{a}\right)$.

Then $\lim_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\log_e a$ is equal to

(A) $\frac{3}{2}+\frac{\pi}{4}$ (B) $\frac{3}{8}+\frac{\pi}{4}$ (C) $\frac{5}{2}+\frac{\pi}{8}$ (D) $\frac{3}{4}+\frac{\pi}{8}$

Ans. (C)

Solution:

$f:(-\infty,\infty)-\{0\}\to R$

$f'(1)=\lim_{a\to\infty}a^2 f\left(\frac{1}{a}\right)$

$\lim_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\ell n(a)$

$\lim_{a\to\infty}a^2\left(\frac{\left(1+\frac{1}{a}\right)}{2}\tan^{-1}\left(\frac{1}{a}\right)+1-\frac{2}{a^2}\ell n(a)\right)$

$f(x)=\frac{1}{2}(1+x)\tan^{-1}(x)+1-2x^2\ell n(x)$

$f'(x)=\frac{1}{2}\left(\frac{1+x}{1+x^2}+\tan^{-1}(x)+4x\ell n(x)\right)+2x$

$f'(1)=\frac{1}{2}\left(1+\frac{\pi}{4}\right)+2$

$f'(1)=\frac{5}{2}+\frac{\pi}{8}$

Illustration 43

The value of $\lim_{x\to0}2\left(\frac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}.....\sqrt[10]{\cos 10x}}{x^2}\right)$ is _____.

Ans. (55)

Solution:

$\lim_{x\to0}2\left(\frac{1-\left(1-\frac{x^2}{2!}\right)\left(1-\frac{4x^2}{2!}\right)\left(1-\frac{9x^2}{2!}\right).....\left(1-\frac{100x^2}{2!}\right)}{x^2}\right)$

By expansion

$\lim_{x\to0}\frac{2\left(1-\left(1-\frac{x^2}{2}\right)\right)\left(1-\frac{1}{2}.\frac{4x^2}{2}\right)\left(1-\frac{1}{3}.\frac{9x^2}{2}\right).....\left(1-\frac{1}{10}.\frac{100x^2}{2}\right)}{x^2}$

$\lim_{x\to0}2\left(\frac{1-\left(1-\frac{x^2}{2}\right)\left(1-\frac{2x^2}{2}\right)\left(1-\frac{3x^2}{2}\right).....\left(1-\frac{10x^2}{2}\right)}{x^2}\right)$

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$$\lim_{x\to 0}\frac{2\left(1-1+x^2\left(\frac{1}{2}+\frac{2}{2}+\frac{3}{2}+.....+\frac{10}{2}\right)\right)}{x^2}$$

$$2\left(\frac{1}{2}+\frac{2}{2}+\frac{3}{2}+.....+\frac{10}{2}\right)$$

$1 + 2 + ... + 10 = \frac{10\times 11}{2} = 55$

Illustration 44

If α, β are the distinct roots of $x^2 + bx + c = 0$, then $\lim_{x\to\beta}\frac{e^{2(x^2+bx+c)}-1-2(x^2+bx+c)}{(x-\beta)^2}$ is equal to:

(A) $b^2 + 4c$ (B) $2(b^2 + 4c)$ (C) $2(b^2 - 4c)$ (D) $b^2 - 4c$

Ans. (C)

Solution:

$$\lim_{x\to\beta}\frac{e^{2(x^2+bx+c)}-1-2(x^2+bx+c)}{(x-\beta)^2}$$

$$\Rightarrow\lim_{x\to\beta}\frac{1\left(1+\frac{2(x^2+bx+c)}{1!}+\frac{2^2(x^2+bx+c)^2}{2!}+...\right)-1-2(x^2+bx+c)}{(x-\beta)^2}\Rightarrow\lim_{x\to\beta}\frac{2(x^2+bx+1)^2}{(x-\beta)^2}$$

$$\Rightarrow\lim_{x\to\beta}\frac{2(x-\alpha)^2(x-\beta)^2}{(x-\beta)^2}$$

$$\Rightarrow 2(\beta-\alpha)^2 = 2(b^2-4c)$$

Illustration 45

$\lim_{x\to 2}\left(\sum_{n=1}^{9}\frac{x}{n(n+1)x^2+2(2n+1)x+4}\right)$ is equal to :

(A) $\frac{9}{44}$ (B) $\frac{5}{24}$ (C) $\frac{1}{5}$ (D) $\frac{7}{36}$

Ans. (A)

Solution:

$$S=\lim_{x\to 2}\sum_{n=1}^{9}\frac{x}{n(n+1)x^2+2(2n+1)x+4}$$

$$S=\sum_{n=1}^{9}\frac{2}{4(n^2+3n+2)}=\frac{1}{2}\sum_{n=1}^{9}\left(\frac{1}{n+1}-\frac{1}{n+2}\right)$$

$$S=\frac{1}{2}\left(\frac{1}{2}-\frac{1}{11}\right)=\frac{9}{44}$$

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